

Assignment 4

Product Thinking & MVP Development

From Idea to Validated Product – Theory into Practice

Course	Software Development Case Study
Week	Week 8-9
Topic	Product Thinking, Ethics & Product Debt
Team Size	Maximum 2 students
Submission	Report (PDF/Word) + Prototype link or file

Objective

This assignment bridges product-thinking theory and hands-on development. You will apply the lecture concepts – MVP, Product-Market Fit, product debt, and ethical responsibility – to your own product idea, producing both a written analysis and a working prototype.

By completing this assignment, you will:

- Formulate a precise MVP hypothesis and define what would prove or disprove it
- Define the PMF signal you will use to determine whether your product is working
- Build a prototype that directly tests your hypothesis
- Apply at least two ethical principles before and during development
- Reflect on what you learned – and on your next hypothesis

CONNECTION TO LECTURE

Your answers must engage with lecture content – the MVP definition, PMF signals, product debt types, Kaspi distribution risk example, Privacy by Design, and IEEE EAD principles. Generic answers that could apply to any product will receive partial credit only.

Key reading references: *The Lean Startup* (Ries, 2011) · *Inspired* (Cagan, 2017) · *The Startup Owner's Manual* (Blank, 2012) · *ACM Code of Ethics* (2018) · *GDPR Arts. 5, 17, 25*

Part 1 – MVP Definition (30%)

This is the most important section of the assignment. Before writing code, you must clearly state what you are building, for whom, and how you will know whether it works.

1.1 MVP Hypothesis (10%)

Define your MVP using the lecture framework: the smallest thing that tests your riskiest assumption about user value.

Field	Your Answer
The problem you are solving	(One sentence – from the user's perspective, not the feature list)
Your riskiest assumption	(What must be true for your product to have value? What would kill it if false?)
Your MVP hypothesis	(Complete the sentence: "If we build [X], we believe [Y users] will [do Z], because [assumption]")
What the MVP is NOT	(What are you explicitly leaving out, and why?)
How you will test it	(What is the minimum you need to build to prove or disprove the hypothesis?)

Eric Ries defines an MVP as a version of a product that enables a full Build-Measure-Learn loop with minimal effort. Your answer should reflect this – not a definition of 'a simple version of the full product.'

1.2 Product-Market Fit Signal (10%)

You will not have PMF yet – that is expected. What you must define now is how you will know when you have it.

Question	Your Answer
Who is your primary user?	(Be specific – not 'everyone' or 'students')
What retention signal will you track?	(Not signups or downloads – what behaviour shows users find real value?)
Distribution risk	(Who could add your core feature as a button in an existing product? Would that kill you? Why or why not?)

The Kaspi bill-splitting case from the lecture is directly relevant to the distribution risk question. Your answer must name actual competitors or platforms in your market.

1.3 Product Debt Awareness (10%)

Identify the product debt risks most likely to affect your product. Product debt (poor decisions about what to build) is distinct from technical debt (shortcuts in how you build it).

Debt Risk	How It Could Appear in Your Product	How You Will Avoid It
Feature creep		
Wrong early-adopter assumption	(Your first users may not represent your real market)	
North star metric misalignment	(Optimising for engagement when value is task completion, for example)	

Part 2 – Ethics & Privacy (20%)

Ethical decisions are architectural decisions. This section must be completed before you write code, not after.

2.1 Privacy by Design (10%)

For each principle, describe how your product will implement it. If a principle does not apply to your product, explain why.

Principle	How Your Product Applies It
Data minimisation – collect only what you use	
Purpose limitation – data collected for one feature cannot be reused for another	
Default = private – users opt in to sharing, not out	
Right to erasure (GDPR Art. 17) – full deletion including logs and backups	

2.2 Ethical Risk Analysis (10%)

Identify two ethical risks specific to your product. Apply the IEEE Ethically Aligned Design (EAD1e, 2019) principles. Generic answers, such as ‘data might be hacked,’ will not receive credit.

Risk	EAD Principle	Who Is Affected	Your Mitigation
Risk 1:			
Risk 2:			

IEEE EAD five principles: (1) Human Well-being, (2) Accountability, (3) Transparency, (4) Awareness of Misuse, (5) Competence. Apply the

Greyball test: if a feature must be hidden from users or regulators to function, it is an ethics violation, not a product decision.

Part 3 – MVP Prototype (40%)

Build the smallest working prototype that directly tests the hypothesis you defined in Part 1.1. You are not building the full product – you are building evidence for or against your riskiest assumption.

3.1 Prototype Requirements

Component	Requirement	Weight
Core Feature	At least one end-to-end workflow that directly tests your MVP hypothesis. Input → Processing → Output. No fake buttons.	25%
User Interface	Minimum 2 screens with functional navigation. UI must reflect your target user.	10%
Data Flow	Data must change based on user input. Mock data is acceptable.	5%

Depth over breadth. One fully implemented feature that proves your hypothesis is worth more than five half-built features that prove nothing.

3.2 Hypothesis Test Report

After building the prototype, answer the following four questions in your written report. This is the most important part of Part 3.

Question	Your Answer
Did the prototype test your MVP hypothesis?	(Reference the exact hypothesis from Part 1.1)
What did you learn that you did not know before building?	(What assumption turned out to be wrong or incomplete?)
What product debt did you introduce, and why?	(Did any of the debt risks from Part 1.3 appear?)
What is the next hypothesis to test?	(The next riskiest assumption your prototype did not address)

Part 4 – Reading Connection (10%)

Select one source from the lecture reading list and connect it directly to a decision you made in Parts 1–3. This is not a summary – it is evidence that you read the source and applied it to your product.

Source (title, author, year)	Key idea you applied	Where it appears in your work

Example: ‘Steve Blank’s customer development methodology (The Startup Owner’s Manual, 2012) led us to interview 5 users before writing code. This changed our MVP hypothesis from [X] to [Y] because users told us [Z].’

Acceptable sources from the lecture reading list:

- The Lean Startup – Eric Ries (2011)
- Inspired – Marty Cagan (2017)
- The Startup Owner’s Manual – Steve Blank (2012)
- ACM Code of Ethics – ACM (2018)
- GDPR Full Text – EU (2018), Arts. 5, 17, 25
- IEEE Ethically Aligned Design, EAD1e – IEEE (2019)
- Weapons of Math Destruction – Cathy O’Neil (2016)

Deliverables & Grading

	Item	Content	Format
1	Written Report	Parts 1, 2, 4 + Hypothesis Test Report (Part 3.2). Recommended: 4–6 pages.	PDF or Word
2	Prototype	Working code. GitHub repo preferred. README with setup instructions required.	GitHub / link / file

Grading Rubric

Section	%	Exemplary	Satisfactory	Needs Improvement
Part 1: MVP Definition	30%			
1.1 MVP Hypothesis	10%	Specific, falsifiable hypothesis. Riskiest assumption clearly identified.	Hypothesis present but vague. Riskiest assumption partially identified.	MVP defined as ‘simple version of full product.’ No

		References lecture framework.		falsifiable hypothesis.
1.2 PMF Signal	10%	Specific retention signal. Distribution risk names real competitors. Kaspi-type analysis applied.	PMF signal defined but generic. Distribution risk mentioned without specifics.	PMF confused with signups or downloads. No distribution risk.
1.3 Product Debt	10%	All three debt types addressed with product-specific examples. Lecture cases referenced.	Two types addressed. Some product-specificity.	Generic or missing. No connection to lecture content.
Part 2: Ethics & Privacy	20%			
2.1 Privacy by Design	10%	All four principles applied with specific decisions. Right to erasure addresses logs and backups.	At least two principles addressed. Some specificity.	Generic checklist. No architectural decisions. GDPR not engaged.
2.2 Ethical Risk Analysis	10%	Two risks with correct EAD principle mapped. Greyball test applied. Mitigation actionable.	Two risks identified. EAD partially applied. Mitigation general.	Generic risks. No EAD framework. No Greyball test.
Part 3: MVP Prototype	40%			
Prototype	30%	Directly tests MVP hypothesis. Core workflow end-to-end. Dynamic data. UI reflects target user.	Prototype works but link to hypothesis is unclear. Basic data flow.	Static mockup or non-functional. No connection to hypothesis.
3.2 Hypothesis Test Report	10%	Honest, specific reflection. New learning articulated. Product debt acknowledged. Next hypothesis defined.	Reflection present but surface-level.	Reflection absent or only confirms the hypothesis without evidence.
Part 4: Reading Connection	10%			
Reading Reflection	10%	One source cited with specific idea applied. Clear link to a product decision made. Evidence of actual reading.	Source cited. Connection to work partial.	Source listed without application, or not from reading list.

Submission Checklist

☐	MVP hypothesis is falsifiable and references the riskiest assumption (not just a feature description)
☐	Distribution risk names at least one real potential competitor or platform in your market
☐	Privacy by Design table includes architectural decisions, not just intentions

☐	At least two ethical risks with an EAD principle mapped to each
☐	Prototype runs without errors and includes a README
☐	Hypothesis Test Report is in the written report (not only in the README)
☐	One reading list source cited with specific application to your product
☐	Report is 4–6 pages, PDF or Word, submitted alongside the prototype link or file

FROM THIS ASSIGNMENT TO THE FINAL PRESENTATION

The MVP hypothesis you define here is the benchmark for your final presentation. At the end of the course, the question will be: did you test what you said you would test, did you build what matters, and did you ship something you stand behind ethically?